YEAR 2 ARCHITECTURE PORTFOLIO



DERIN ENHOS

project 1

malesa daycare refurbishment



Shadow and Sun Analysis - On Site

West

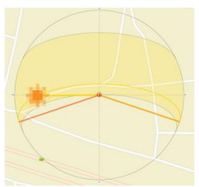


Sun Setting

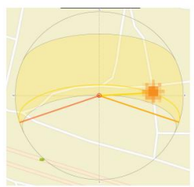


Sun Rising

East



3pm (typical school day end)



7am (typical school day start)

Site Photos - Malesa Daycare















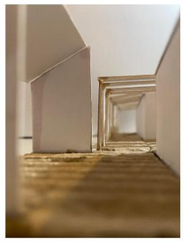
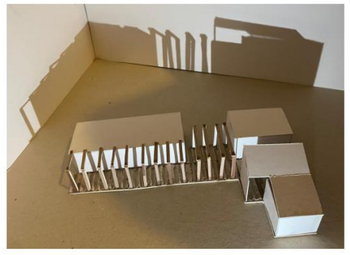


site analysis

Design Process - 1:100 Model

Based off my initial floor plan and heavily inspired by the repetitive collumns from my precedent 'The Stoa of Attalos', I experimented with creating a repetitive pattern that would guide a path through the site from the entry to the main building.



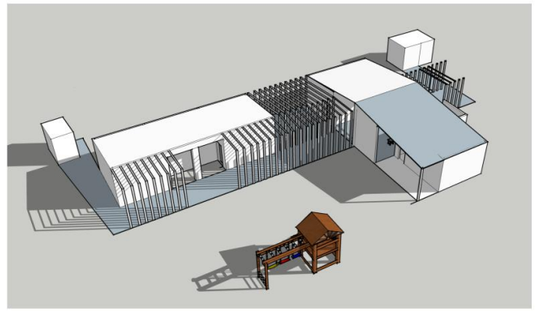
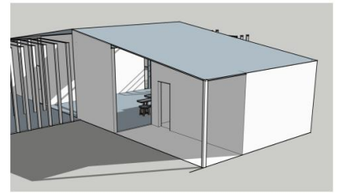
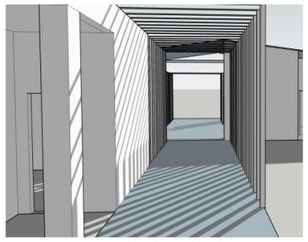
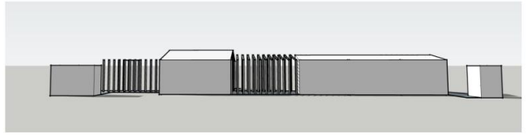
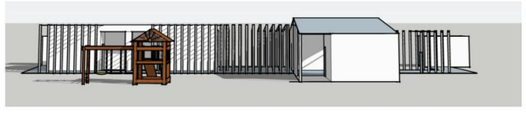
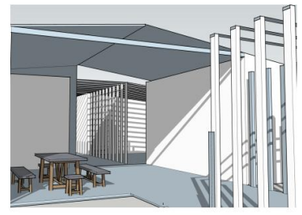

Experimenting with creating playful shadows





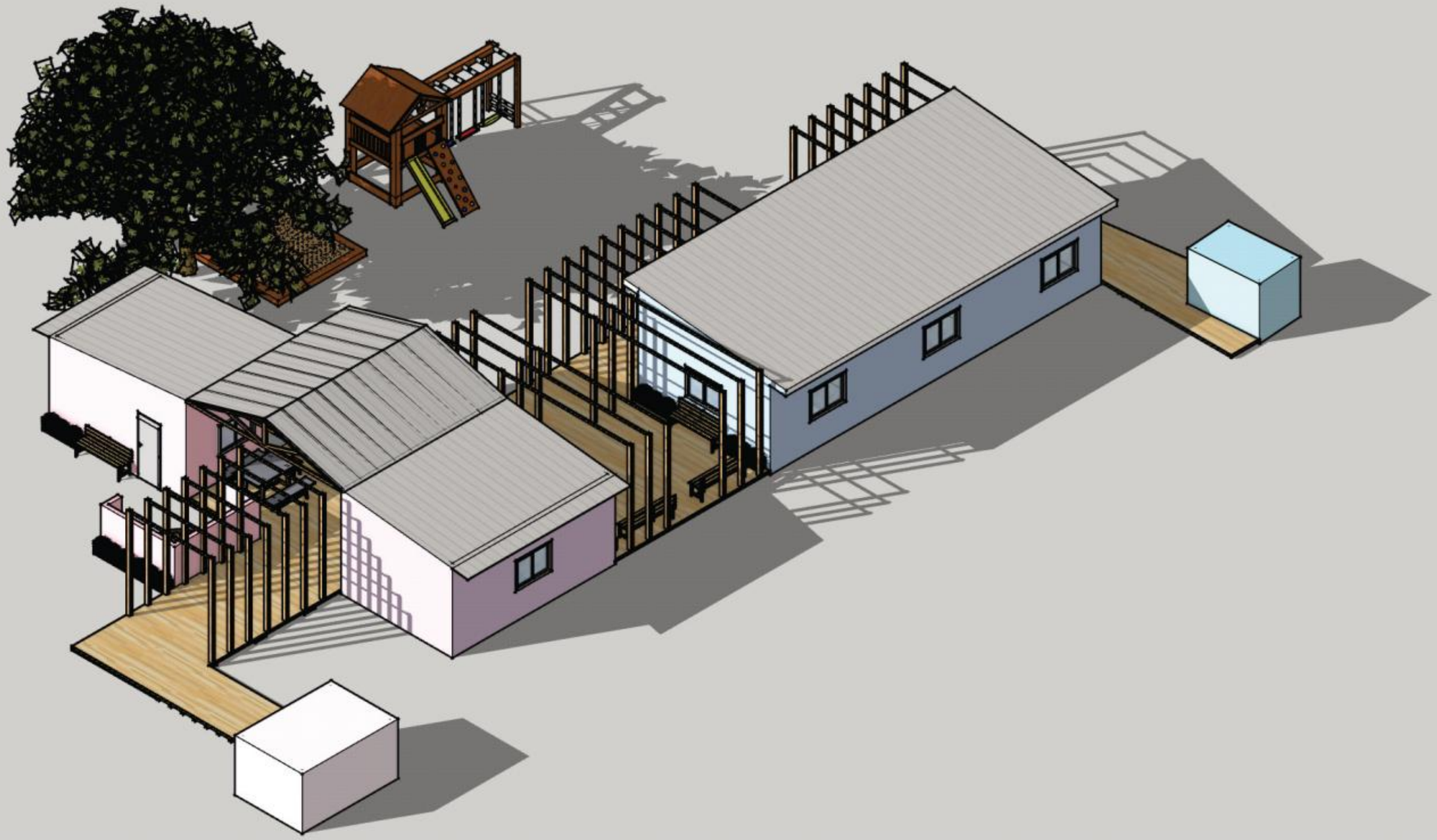


Design Process - 3D SketchUp Model

the design process

Design Proposal - Axonometric Model 1:100



existing buildings

new buildings

timber deck

vegetation

0 1 2 3 4m

1:100

N

proposal floor plan

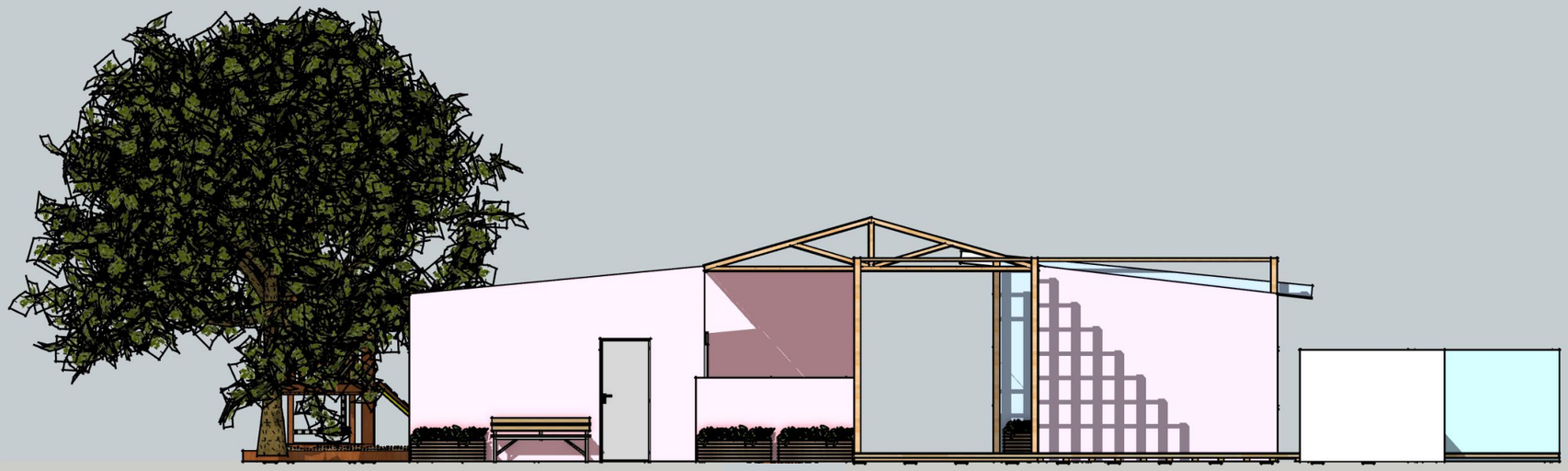


roof plan

elevations



west elevation 0 1 2 3 4m 1:100 N



north elevation 0 1 2 3 4m 1:100 N



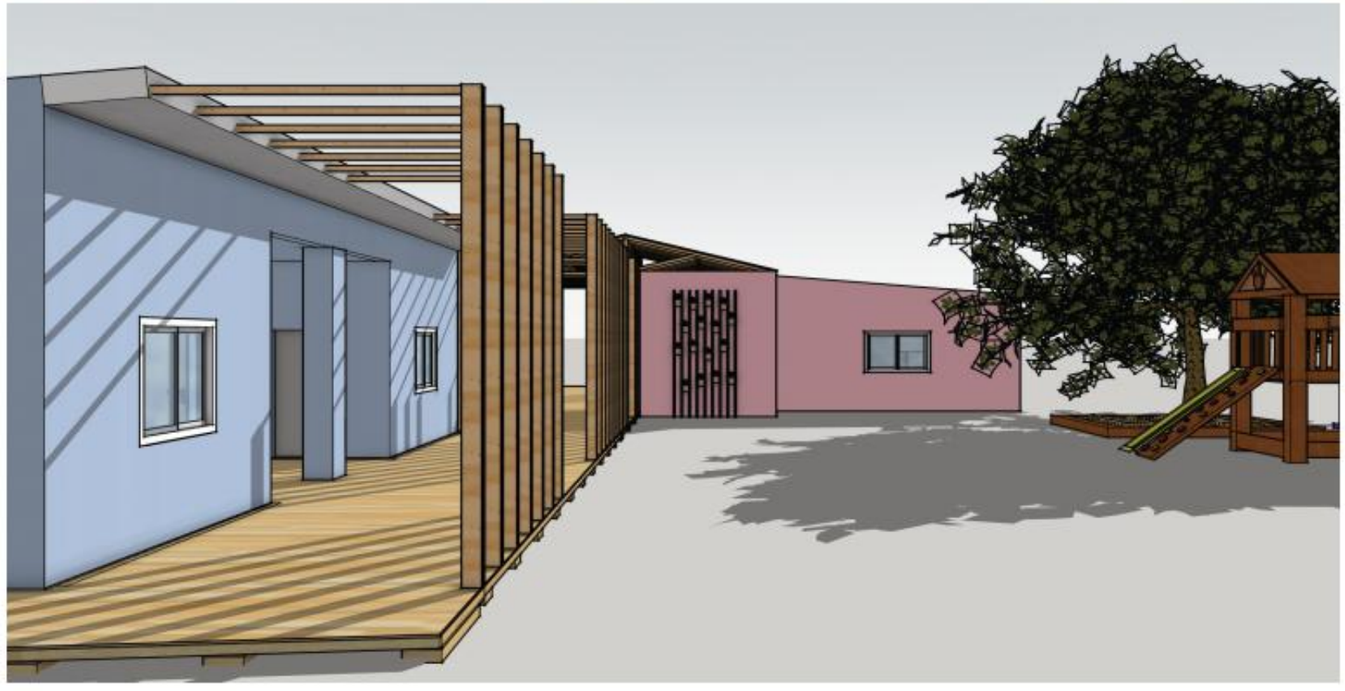
east elevation 0 1 2 3 4m 1:100 N

Design Proposal - 3D Views





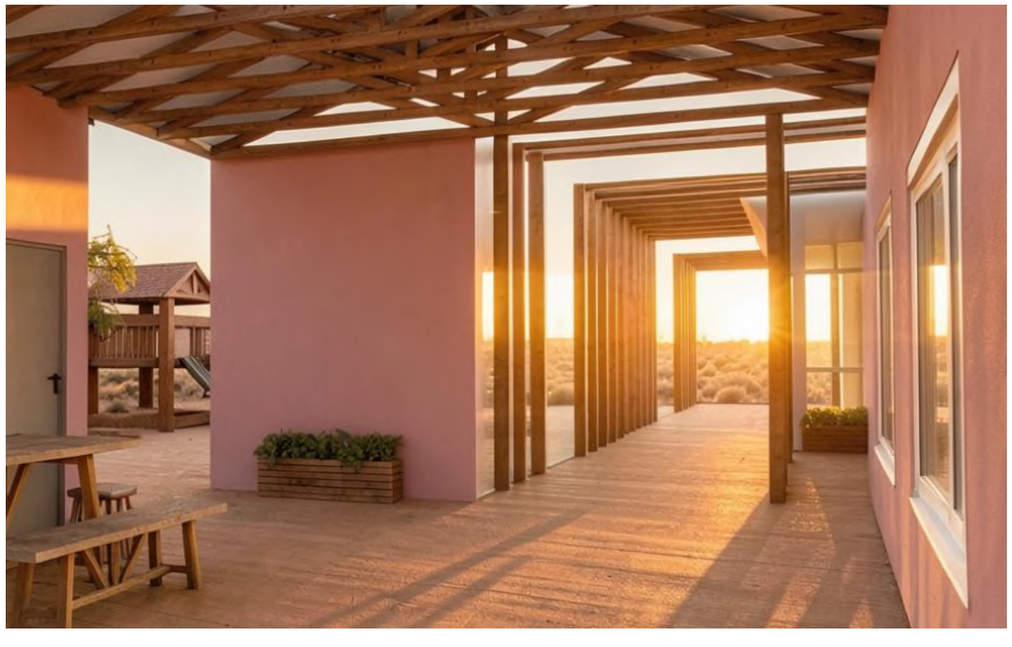












rendered views



project 2

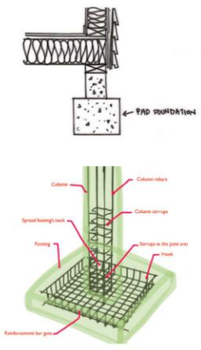
modular preschool construction in south africa



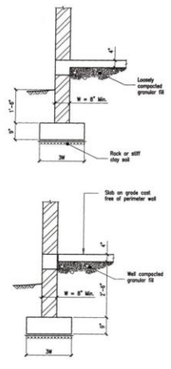
The System - Foundation details and types

My system works in a way where it is suitable to be placed on these 2 different types of floors, supported by these 3 different types of foundations. This enables my system to be placed on more than one type of site, meaning it is adaptable for many sites in South Africa.

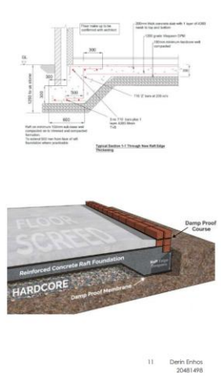
SUSPENDED TIMBER FLOOR WITH PAD FOUNDATION



SLAB FOUNDATION

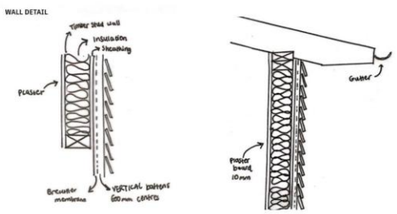


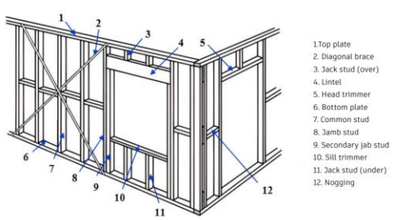
RAFT FOUNDATION



The System - Details

WALL DETAIL





1.Top plate
2. Diagonal brace
3. Jack stud (over)
4. Lintel
5. Head trimmer
6. Bottom plate
7. Common stud
8. Jamb stud
9. Secondary jab stud
10. Sill trimmer
11. Jack stud (under)
12. Nogging



Timber Frame + Modular - Chosen system

Advantages:
1. Cost effective
2. Transportable
3. Practical
4. Light weight structure
5. Sustainabilty
6. Low skill labour

Disadvantages
1. Not very customisable
2. Requires a crane or hi-ab lorry
3. Fixed sizes
4. Fire resistance concerns
5. Moisture sensitivity

Post and beam construction

Typical dimensional information:

Structural Timber	Structural Beams	Brandering
38 x 76 S5 PAR 0.9 to 6.6	50 x 152 Rough Sawn 3.0 to 6.6	38 x 38 BB 3.0 to 6.6
38 x 114 S5 PAR 0.9 to 6.6	50 x 228 Rough Sawn 4.2 to 6.6	
38 x 114 S7 PAR 4.2 to 6.6	38 x 152 S7 PAR 4.2 to 6.6	38 x 228 S7 PAR 4.2 to 6.6
38 x 152 S5 PAR 3.0 to 6.6	50 x 76 S5 PAR 3.0 6.6	38 x 228 S5 PAR 3.0 to 6.6

Fire Performance: Charring Rate: Timber forms a protective char layer when exposed to fire, slowing the burn rate (softwood: ~0.8 mm/min, hardwood: ~0.6 mm/min).

Fire Resistance: Engineered timber often meets higher fire safety standards due to treatments.

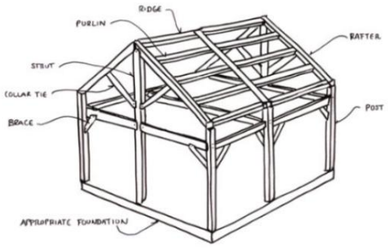
Carbon Impact: Timber is a low-carbon material as trees absorb CO_2 during growth. Acts as a carbon store throughout its lifespan, provided it isn't burned or decomposed at the end-of-life.

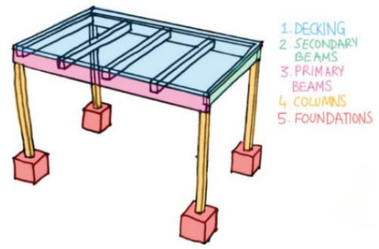
Reusability at End-of-Life

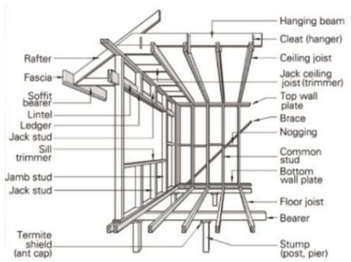
Re-use: timber can often be deconstructed and reused in construction, furniture, or smaller projects, provided it is in good condition. Engineered products like CLT panels can also be repurposed for structural or non-structural applications.

Recycling: Timber waste can be chipped for particleboard, mulch, or bioenergy. Treated timber may have limitations in recycling due to chemicals.

End-of-Life Carbon Impact: Reuse and recycling reduce emissions compared to landfilling or incineration. If burned, timber releases stored carbon, contributing to emissions unless used for energy recovery.



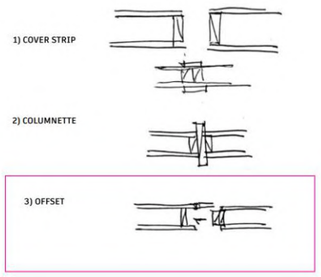




researching timber frame modules + post&beam construction

The System- Connections

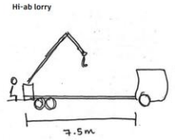
Brainstorming how these cut up modules may connect. Chosen solution marked in **pink**



The Modules - Calculating sizes of modules

My idea is that the building's walls will be modular, they will be built off site and brought in with a Hi-ab lorry . They will come in different sizes in the form of corners and wall panels. This means that anything done on site will be relatively lower-skill labour and it will be more time efficient.

Hi-ab lorry

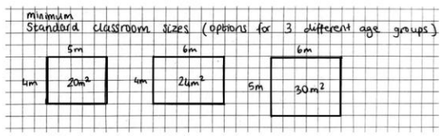


Standard lorry available space:
Length: 7.5m x Width: 2.5m

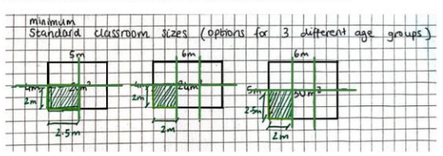
Therefore the modules cannot exceed a size of 2.5 in width.

I chose the maximum measurements as 2.5x2.5 due to weight reasons as the module will be too heavy to transport if it reaches up to 7.5m in length.

This way more than one module can be transported on site at a time saving both time and money.

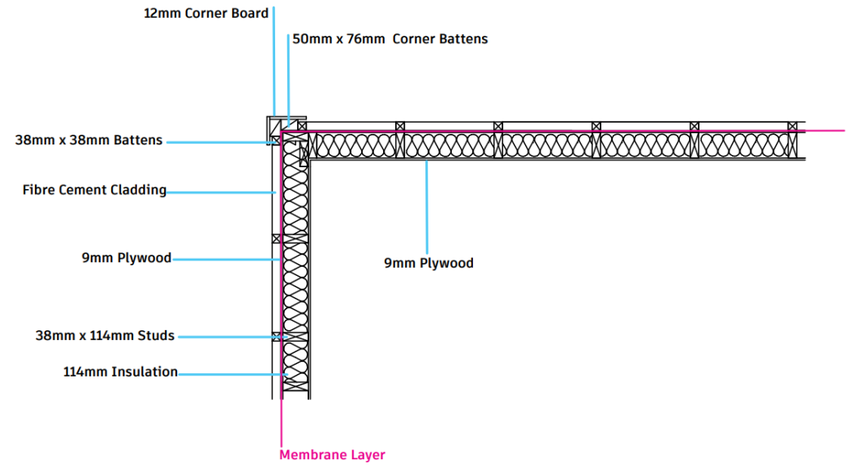


Marked in **green** below is initial dimensions the classrooms can be cut into to be modular blocks.



Details - The Modules - **PLAN VIEW SCALE 1:10**

Here is my modular corner piece that I've designed to be a part of what I call my design 'catalogue' which features the modular pieces that can join to create complete structures.



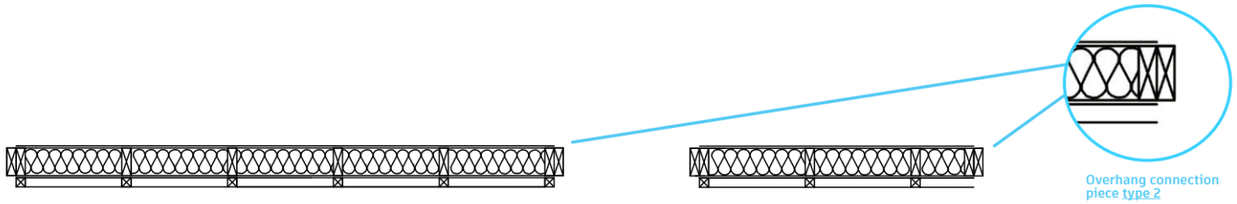
the modular system

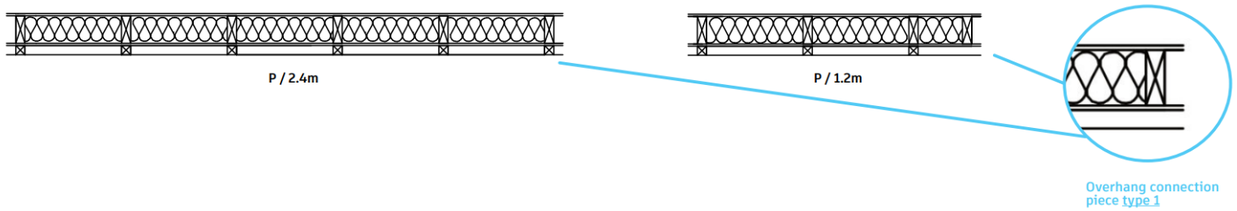
catalogue of modules

Catalogue - The Modules - PLAN VIEW SCALE 1:10

According to the available structural timber joist sizes of **38mm x 114mm** and according to available window widths:
600mm, 900mm, 1200mm
I have decided on the final available sizes shown in this catalogue shown in plan view:

Key:
C = corner
RH = right hand side
T = top
B = bottom
P = panel

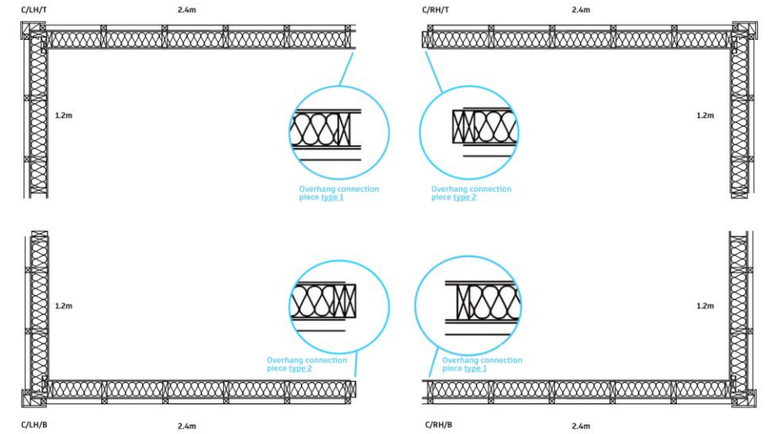




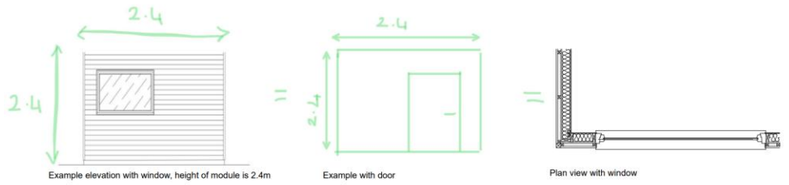
Catalogue - The Modules - PLAN VIEW SCALE 1:10

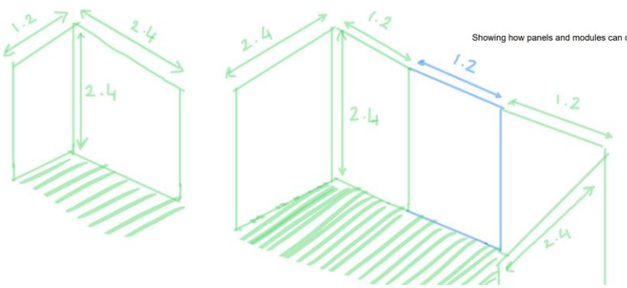
According to the available structural timber joist sizes of **38mm x 114mm** and according to available window widths:
600mm, 900mm, 1200mm
The Modules will be 2.4m tall.
I have decided on the final available sizes shown in this catalogue shown in plan view:

Key:
C = corner
RH = right hand side
T = top
B = bottom
P = panel



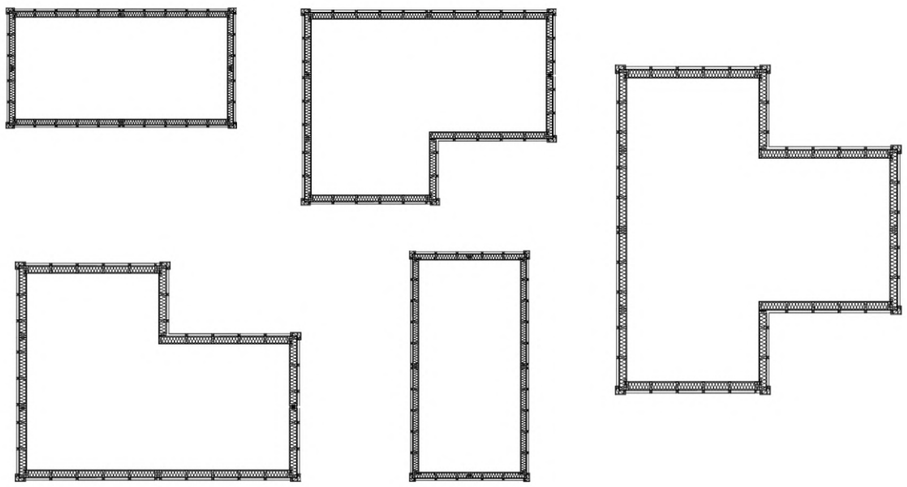
Catalogue - Elevation Sketch of Modules





Connecting - Example connections - **Not to Scale**

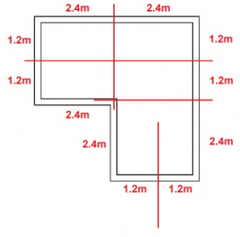
Here are some examples of how the 1.2m x 2.4m corner pieces and the 1.2m and 2.4m panels can be connected and create different combinations of joinery.
There are many posibilities, the examples only depict a few variations.



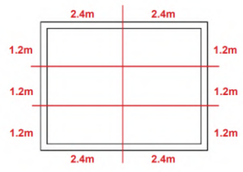
The Design Process - Classroom sizes - **Scale 1:50**

Here I finalised the shapes and sizes of the classroom blocks and marked in red show how these blocks are made up of joint modules from my 'catalogue'.

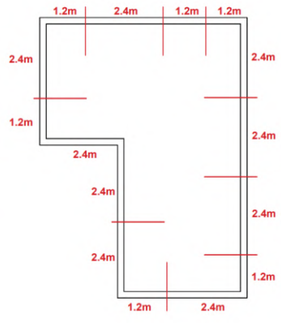
Kitchen + Sick Bay = 17.28m^2



Classroom 1 + Office = 17.28m^2

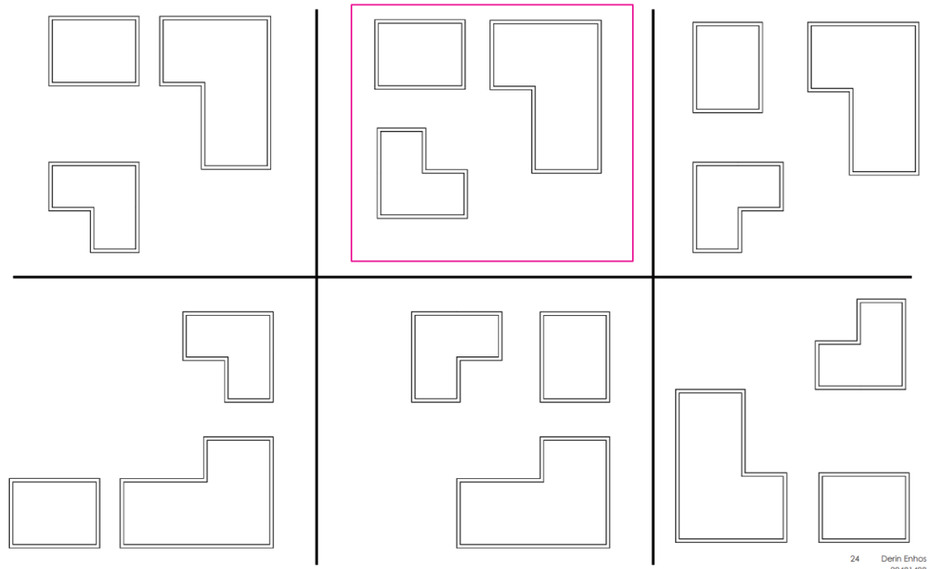


Classroom 2 + 3 = 38.88m^2



The Design Process - Iterations

Here I was playing around with different layouts, the relation between blocks and the creation of outdoor courtyards to find a final layout to be put into a scaled floor plan, chosen iteration marked in **pink**.



Approximate Weight Calculations

Materials from the Drawing:
- Timber Frame: 38mm × 114mm studs
- Sheathing: 9mm plywood
- Cladding: Fibre cement cladding
- Battens: 38mm × 38mm & 50mm × 76mm corner battens
- Insulation: 114mm (assumed mineral wool or PIR foam)

Structure Dimensions:
- Width: 2.4m
- Depth: 1.2m
- Height: 2.4m

Step 1: Timber Frame Weight
C24 softwood (38mm × 114mm) weighs ~3.1kg per meter
Frame estimate (~18m total length of studs, plates, and extra supports):
18m × 3.1kg/m = ~56kg

Step 2: Plywood Sheathing
9mm plywood density: ~6.5kg/m^2
Total area (~14.4m^2 covering inside and outside):
14.4m^2 × 6.5kg/m^2 = ~94kg

Step 3: Fibre Cement Cladding
Density varies, but typically ~11-15kg/m^2
Estimated area (~7.2m^2 on exterior-facing side):
7.2m^2 × 13kg/m^2 = ~94kg

Step 4: Battens and Corner Boards
38mm × 38mm battens (~8m total): ~15kg
50mm × 76mm corner battens (~4m total): ~15kg
12mm corner board (~2m^2): ~8kg

Step 5: Insulation (114mm thick)
Mineral wool (50kg/m^3) or PIR (~32kg/m^3)
Volume (~2.4m × 1.2m × 0.114m × 2 walls = ~0.65m^3)
Mineral wool: ~32kg
PIR: ~21kg
Estimate: 20-30kg

Final Estimated Weight Summary

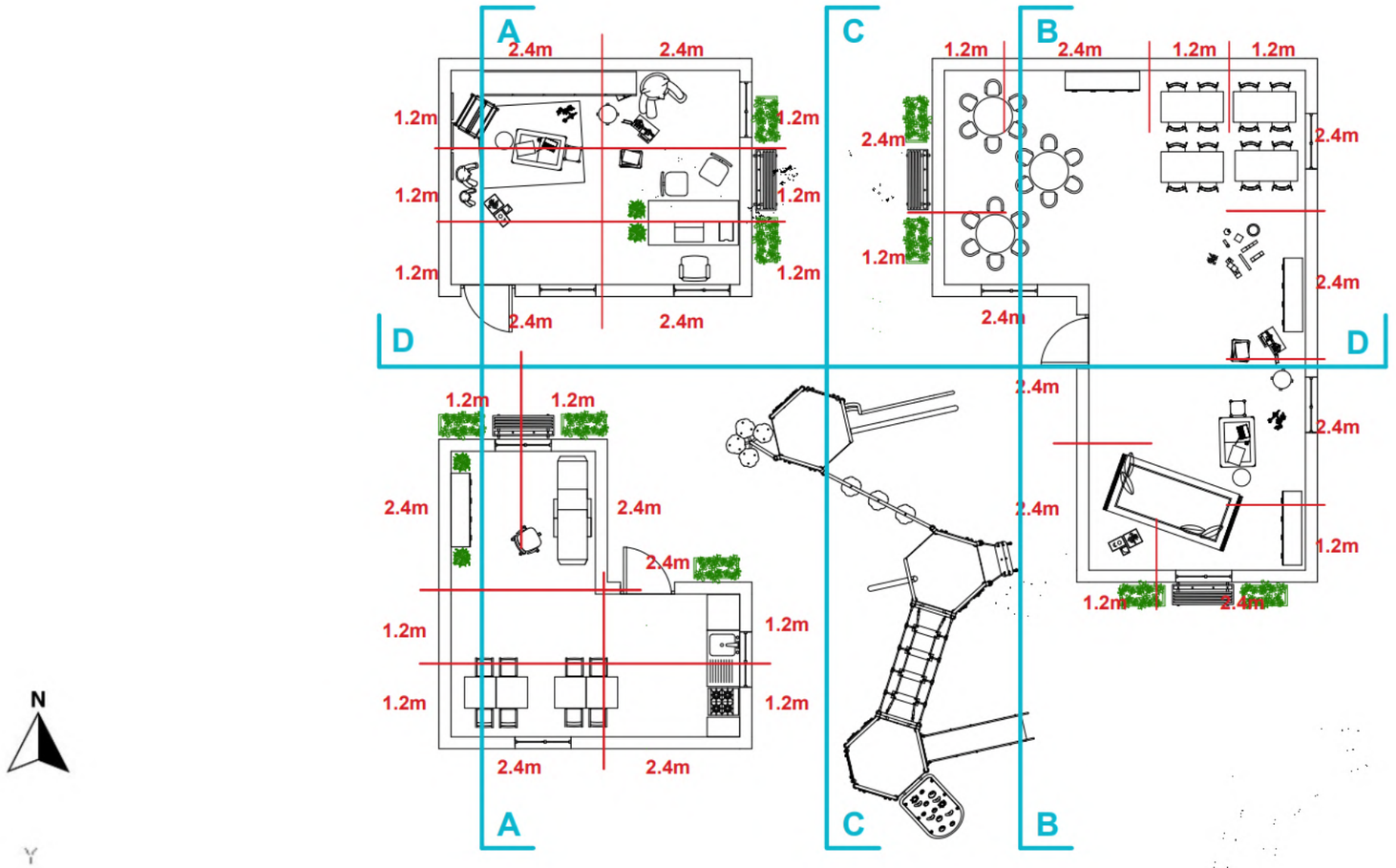
Component	Weight (kg)
Timber Frame	~56
9mm Plywood	~94
Fibre Cement Cladding	~94
Battens + Corner Boards	~38
Insulation	~20-30
Total Estimate	~300-310

Hi - ab Lorry lifting capacity
Small HIAB (Light-Duty, 3-10 Tonne Meter) 500kg - 2,000kg ~6-8m Reach
Medium HIAB (10-30 Tonne Meter) 2,000kg - 8,000kg ~10-15m Reach
Large HIAB (Heavy-Duty, 30-100+ Tonne Meter) 8,000kg - 30,000kg+ ~15-30m Reach

the design process + calculations

Initial Floor Plan - Showing modular blocks in **red**

+ Showing sections and elevations in blue

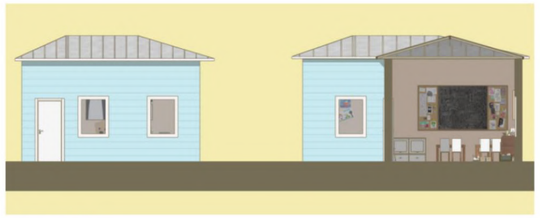


elevations + sections

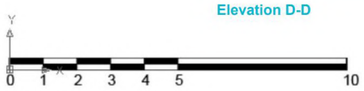
Initial Elevations - Elevations 1:50



Elevation C-C

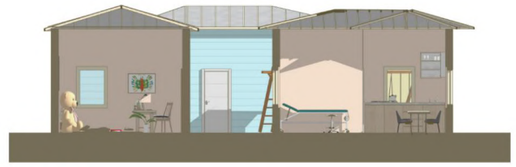


Elevation D-D



SCALE BAR 1:50

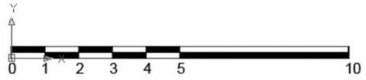
Initial Sections - Sections 1:50



Section A-A



Section B-B



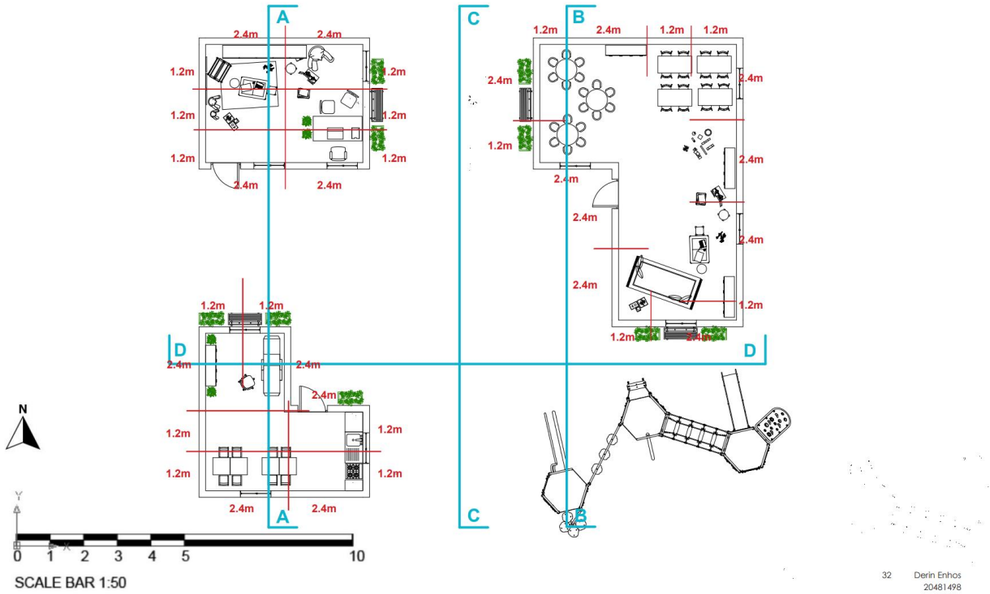
SCALE BAR 1:50

FINAL PROPOSAL

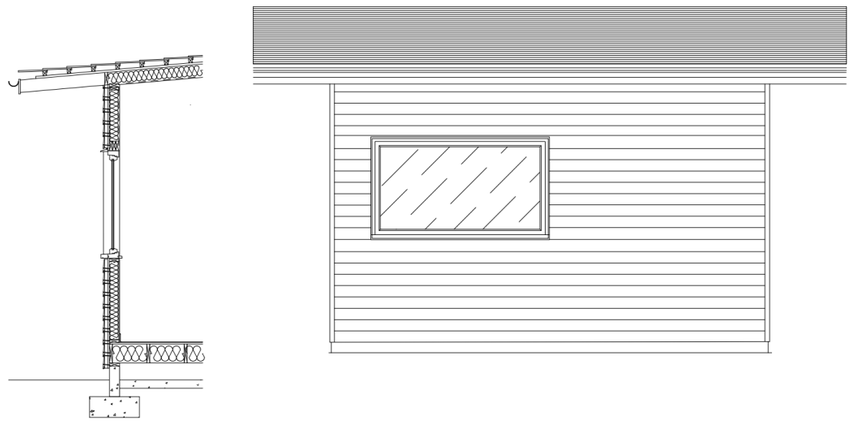
proposal floor plan + details

Initial Floor Plan - Showing modular blocks in **red**

+ Showing sections and elevations in blue

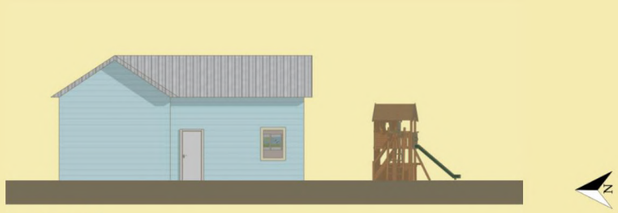


Detailed Section + Elevation - Scale 1:20

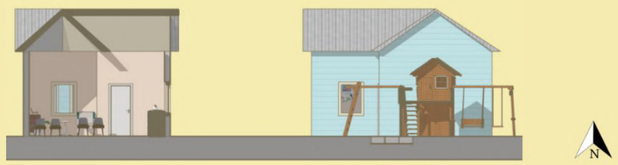


sections + elevations

Initial Elevations - Elevations 1:50



Elevation C-C

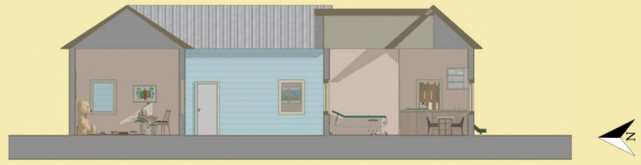


Elevation D-D

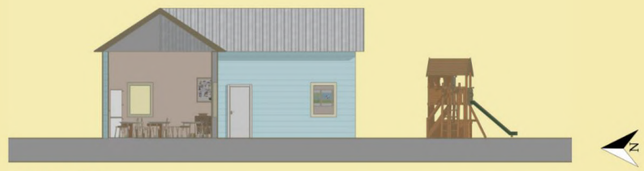


Scale Bar 1:50

Initial Sections - Sections 1:50



Section A-A



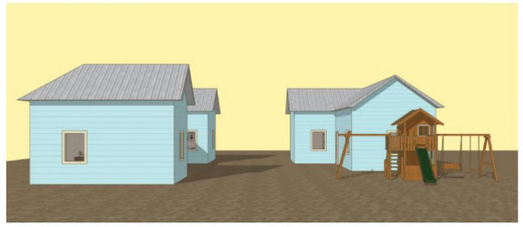
Section B-B



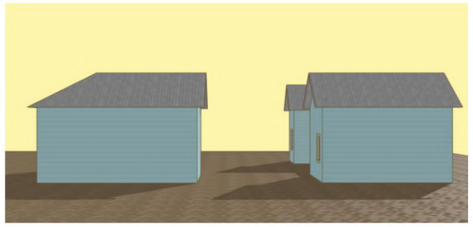
Scale Bar 1:50

Exterior Views - 3D Views

Since the initial design, I've changed the roof from a hip roof with a low pitch to a gable roof with a higher pitch of 1.5m (ridge height) as it is more aesthetically pleasing, the roof will be trusses resting on the modules.



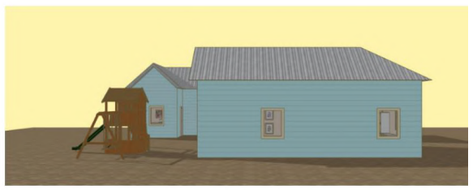
North view



South view



West view



East view

Interior Views - 3D Views



Oldest Age Group Classroom



Kitchen Eating Area



Youngest Age Group Playroom/Classroom



Middle Age Group Classroom Joint



Kitchen



Entrance to Playroom



Oldest Age Group Classroom



Sick Bay



Office in Youngest Age Group Classroom

sketchup views



project 3

community residence + market



GROUND FLOOR PLAN



FIRST FLOOR PLAN



SECOND FLOOR PLAN

Iterations - Analysis of Floor Plans

Here I highlight bits of my floor plans that I do not like or I think does not work. These are things I will be changing going forwards with my design.

Overall comments: Create more relation with the outdoors and do not have residential units on the ground floor for more privacy. More communal or rentable spaces needed, more greenery and access to roof. Think back to concept and relate my design to it.



Ground floor plan



Second floor plan

the design process

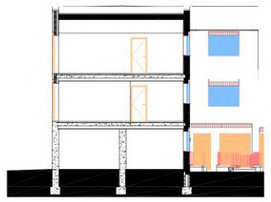

FIRST
FLOOR
PLAN

SECOND
FLOOR
PLAN

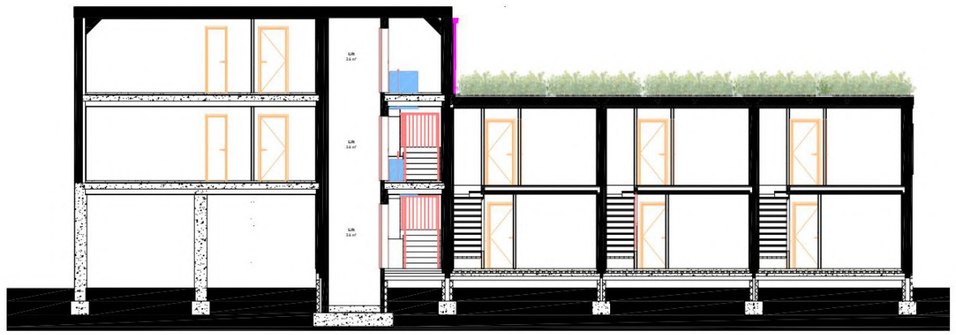

GROUND
FLOOR
PLAN
Sliding glass door
Workshop 1
Workshop 2
Open to outdoor Workshop 3
Access by truck
existing db
N
0 1 2 3 4 5 6 7 8 9 10

proposal

SECTION A-A
SCALE: 1:50



SECTION B-B
SCALE: 1:50



SCALE BAR 1:50

SECTION C-C
SCALE: 1:50



sections (autocad)

elevations

SOUTH ELEVATION



NORTH ELEVATION



EAST ELEVATION



WEST ELEVATION



sketchup model

3D PROJECTION



3D PROJECTION



3D PROJECTION



3D VIEWS



Main Entrance



Rentable Stalls/Market



Open to Outdoor Workshop



Green Roof with Roof Access



tectonics



Part 1 - Structural Schematic

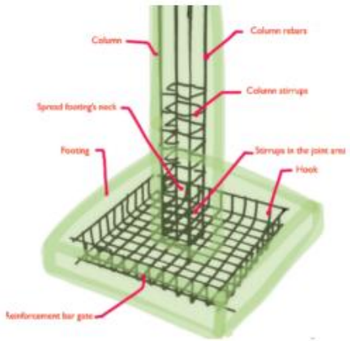
This project employs a hybrid structural system that integrates softwood timber framing, loadbearing masonry, light steel components, and reinforced concrete foundations. Engineered timber-steel Posi-Joists are used for the intermediate floor, introducing light steel into the timber-dominated structure. This approach creates an efficient, cost-effective, and buildable system with excellent thermal and acoustic performance, flexible service voids, and high insulation standards.

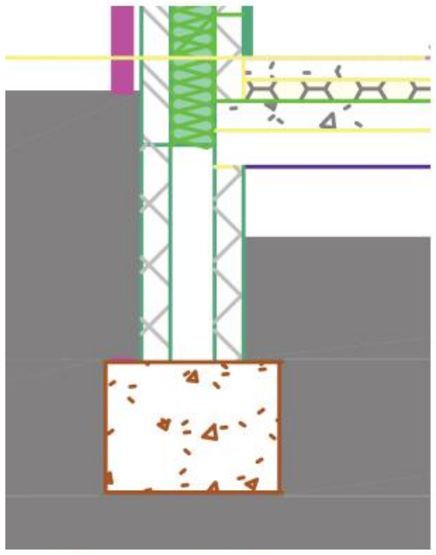
Foundation System

Type: Pad foundation with 450mm deep reinforced concrete footings.

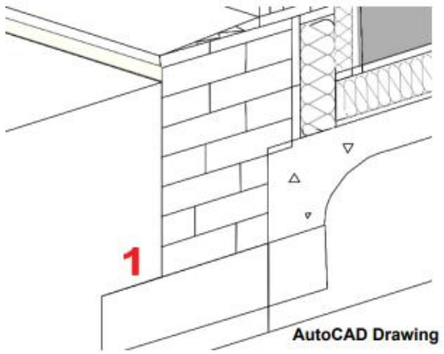
Function: Distributes structural loads from the building into the ground and resists settlement.

Materials: Concrete with reinforcement bars.





Screenshot taken from Section D-D



AutoCAD Drawing

External Walls

Type: Loadbearing timber stud wall with insulation, brick cladding.

Build-up:

Facing brickwork (non-structural outer leaf)

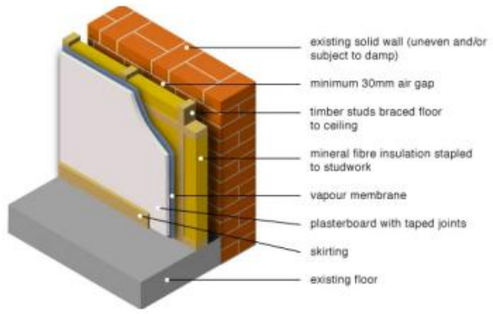
Breather membrane

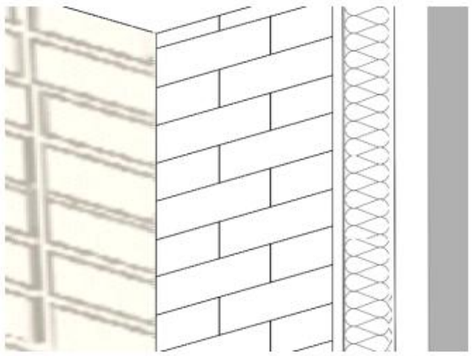
45x145mm softwood timber studs (structural frame)

Wood fibre insulation between studs

Internal insulated plasterboard (with 50mm PIR and 12mm plasterboard)

Function: Provides lateral and vertical support; forms the building envelope with thermal and moisture protection.





AutoCAD Drawing



Intermediate Floor (Hybrid Timber & Steel System)

System: Posi-Joists, engineered joists with timber top and bottom chords and galvanised steel webbing.

Supported by: Internal or external loadbearing timber walls.

Layers:

38x89mm timber studs (support frame)

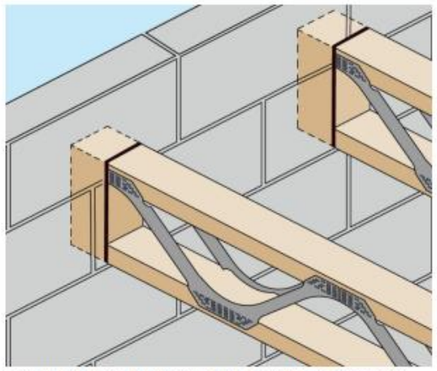
159x253mm Posi-Joists at 600mm centres

Strongbacks for structural bracing

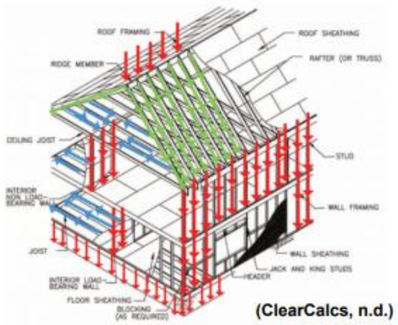
100mm wood fibre insulation between joists

22mm chipboard + cement particle board (floor decking)

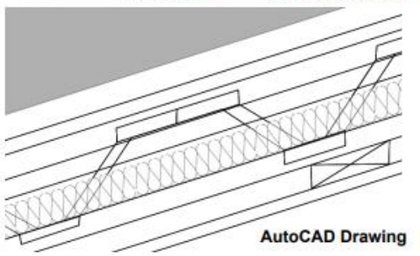
Function: Spans between walls, supports upper floors, and allows flexible routing of services.



(The Posi-Joist Technical Handbook Issue 7, n.d.)



(ClearCalcs, n.d.)



AutoCAD Drawing



Ground Floor

Build-up:

Limecrete slab on aggregate

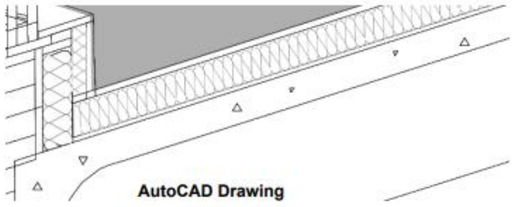
Liquid-applied DPM and vapour control membrane

150mm foam glass insulation (Glapor)

5mm foil-backed separating layer

75mm anhydrite screed

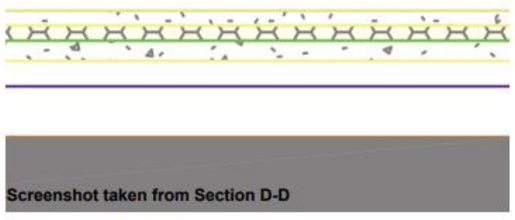
Function: Thermally efficient and breathable ground-bearing slab system.



AutoCAD Drawing



(Great Home, 2020)



Screenshot taken from Section D-D

Roof Structure (Timber)

Structure: LVL (laminated veneer lumber) rafters (47x150mm) with wood fibre insulation between.

Layers:

Internal gypsum plasterboard ceiling

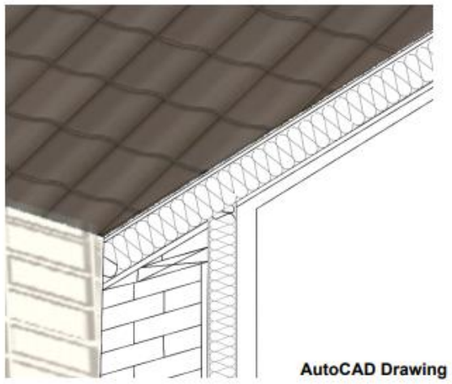
PIR insulation and vapour control layer

Tyvek breathable membrane

Vertical + counter battens

Clay tiles

Function: Transfers roof loads to walls, provides weatherproofing, and contributes to thermal envelope.



AutoCAD Drawing

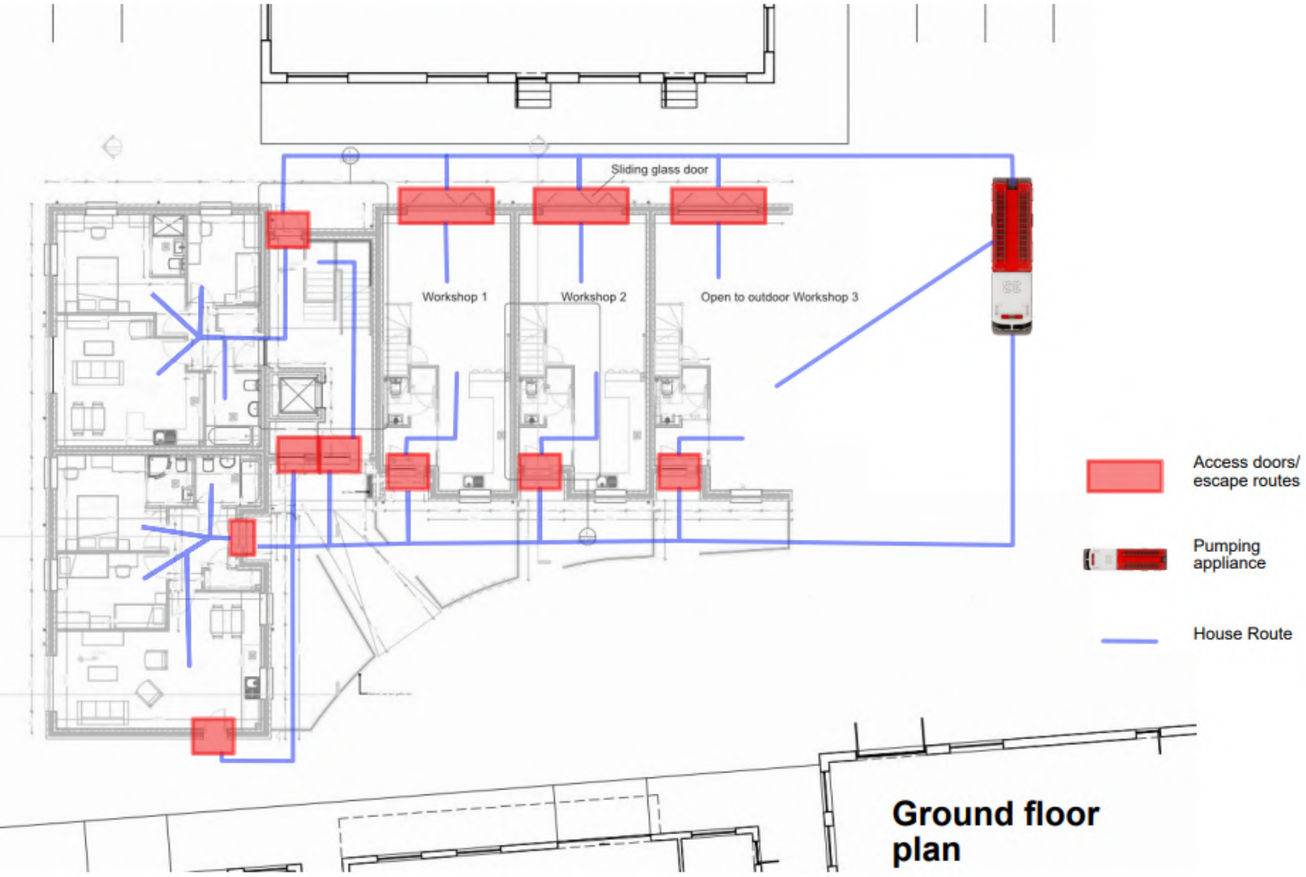
Structural Load Path Overview

Roof loads (tiles, rafters, insulation) → transferred through LVL rafters → to external walls.

Intermediate floor loads → transferred via Posi-Joists → to internal/external loadbearing walls.

Wall loads → transmitted vertically down to reinforced concrete pad foundations.

Part 2 - Fire - Strategic Decisions

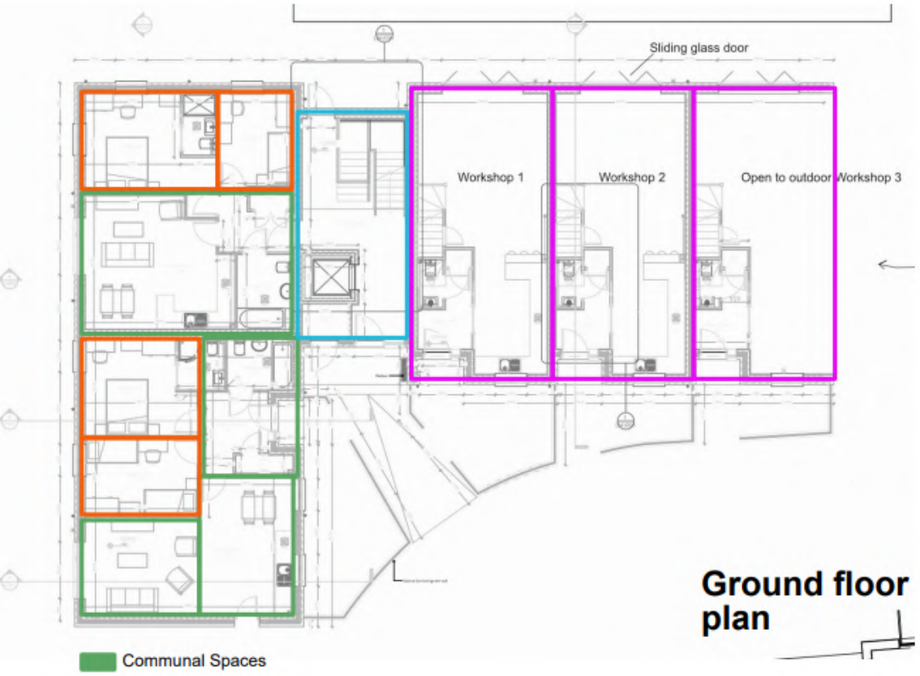


Ground floor plan



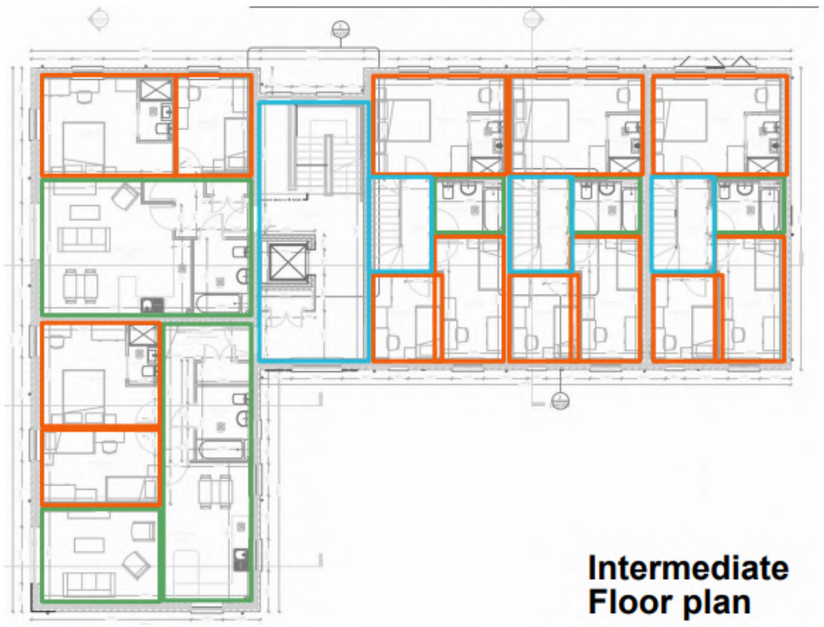
Section C-C

Flats
Stairs
Workshop
Lift



Ground floor plan

Communal Spaces
Dorms
Compartment floors
Workshops



Intermediate Floor plan

Communal Spaces
Dorms
Compartment floors

Section 10 of Approved Document B4 talks about how to prevent the spread of fire due to external walls.

The standards include: Material Combustibility (Regulation 7(2)) where buildings taller than 11m must have noncombustible external walls (Class A2-s1, d0 or better) and that balconies and façades must also meet this requirement. My building has no balconies and is below 11m in height.

Fire-Resistant Materials: The building has a brick external envelope, which is non-combustible and should meet A2-s1, d0 classification.

Inset Window Design: The deeply recessed windows reduce the direct spread of flames between floors.

Compartmentation compliance: The building meets the fire resistance and separation requirements.

Sprinkler compliance: Ther requirement is that if the building has flats taller than 7.5 m in height, it needs sprinklers, the building does have flats at 7.5m in height and therefore has sprinkler system installed, complying with B3 Section 7.

Protected Stairwell: The building's central core contains a single, fire-rated stairwell. There is also stairs in every workshop which provides a means of escape from the intermediate floor flats through the workshop.

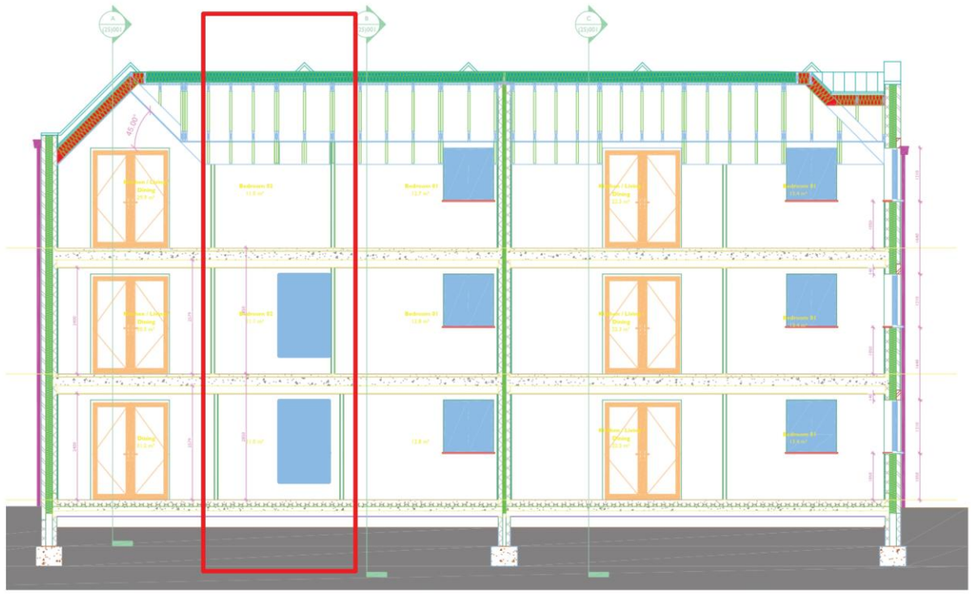
Smoke Ventilation: According to the updated Section 22 of BS 9991, natural ventilation is a suitable smoke ventilation method for buildings up to 30m above street level. Since this building is way below 30m in height, smoke ventilation or pressure differential system is not required.

This building lacks a protected corridor but includes a protected firefighting lobby with firefighting stairs (800mm wide), which serve as a sufficient escape route. Escape stairs must be constructed from A2-s3, d2 or better class materials. For it to comply with Document B, the firefighting shaft must meet the BS 9999 standards.

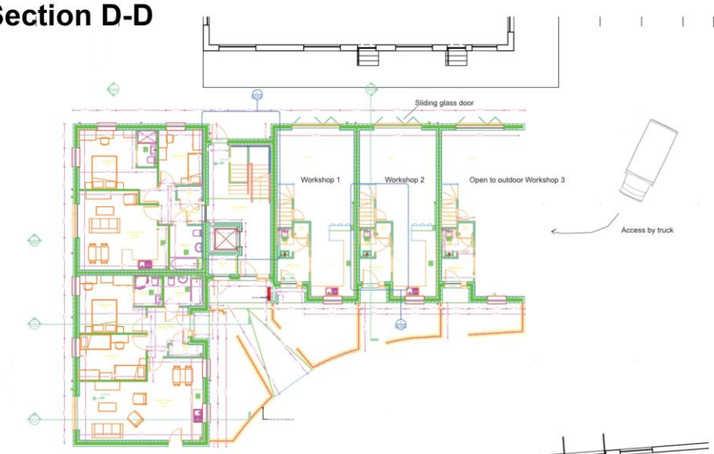
As written on Document B, Section 15.8, lift floors must be within 7.5m of firefighting stairs. In this case, they are located in the same space, meaning it complies. However, it is recommended that the firefighting lobby or corridor be a separate protected space from the firefighting lift shaft and stairs.

Additionally, the firefighting lift is connected to a dry fire main with an outlet valve on each storey, in accordance with Document B, Section 15.10.

Part 3 - Architecture in Detail



Section D-D



Ground Floor Plan



First Floor Plan

1. Foundation

Pad foundation with 450mm deep reinforced concrete footing

2. External wall

Facing brickwork

Breather membrane

45mm x 145mm softwood studs with wood fibre insulation in between

1200mm x 2400mmm Insulated plasterboard (50mm PIR insulation with 12mm plasterboard

3. Window

3mm EDPM rubber flashing

150mm x 180mm concrete cill

DPC lapped under concrete cill

Double glazing

Timber window frame

Thermally broken window fixing with insulated cavity closer and airtight tapes

Foam tape placed under the window frame

Re-formed insulated cavity closers with integrated DPC and fire barrier

4. Internal wall

Facing brickwork

12mm of plaster on both sides of the wall

5. Ground floor

Limecrete slab on aggregate

Liquid-applied DPM; Self-adhesive membrane; Vapour control layer with taped joints

150mm foam glass insulation (Glapor)

5mm Foil-backed separating layer; Vapour control membrane

75mm Anhydrite screed

6. Intermediate floor

12mm plasterboard

38mm x 89mm timber studs

159mm x 253mm Posi-Joists at 600mm centres

3.1mm x 75mm strongback with timber blocks

100mm wood fibre insulation placed between the Posi joists

22mm chipboard

Cement particle board

7. Pitched roof

15mm Gypsum plasterboard

25mm of PIR insulation with 15mm plasterboard

Vapour control membrane with taped seams

Insulated timber battens between external wall and roof

LVL rafters 47mm x 150mm with wood fibre insulation placed in between

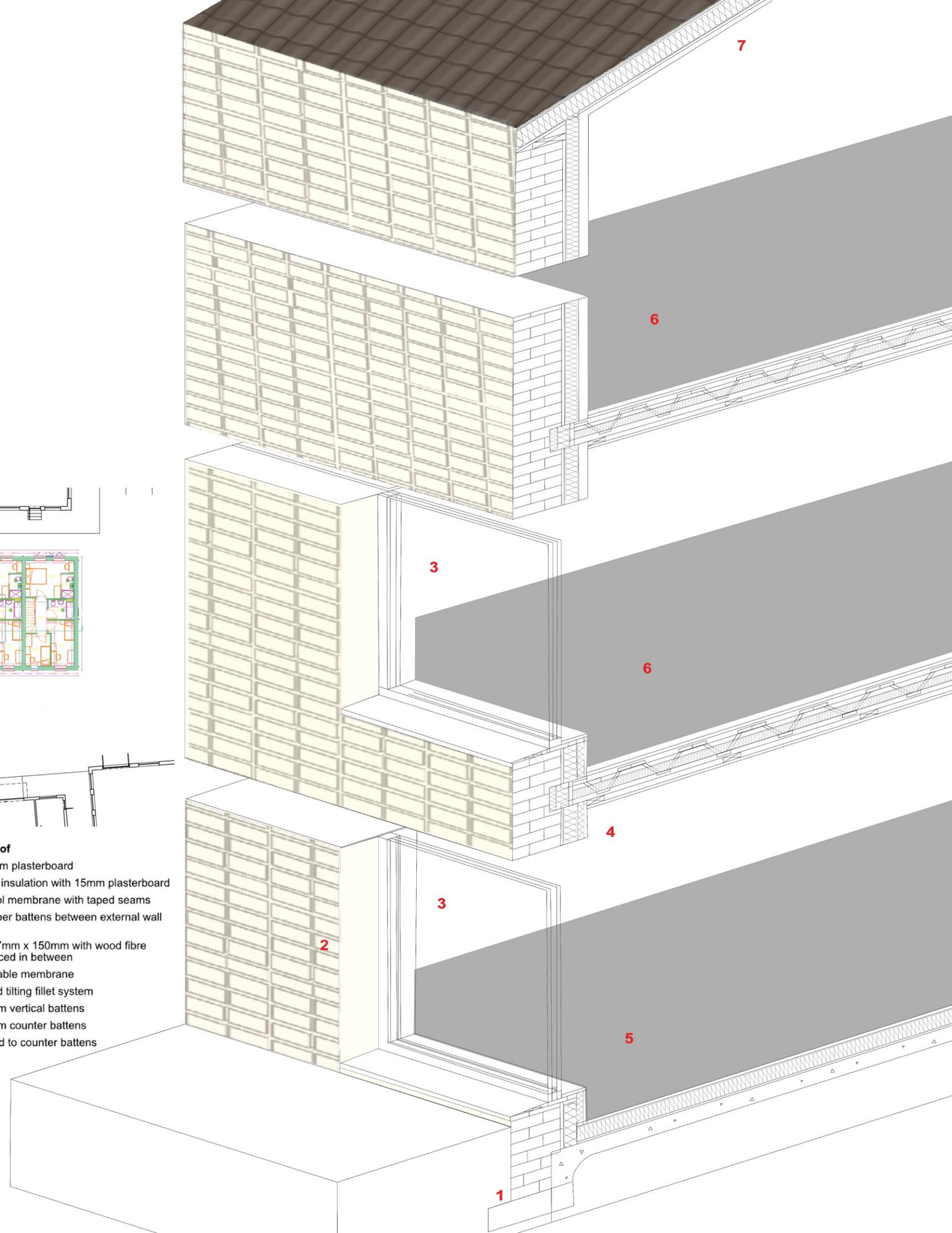
Tyvek breathable membrane

Pre-fabricated tilting fillet system

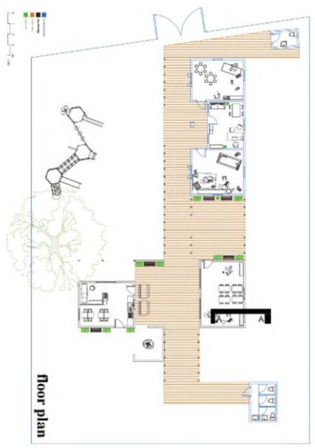
25mm x 50mm vertical battens

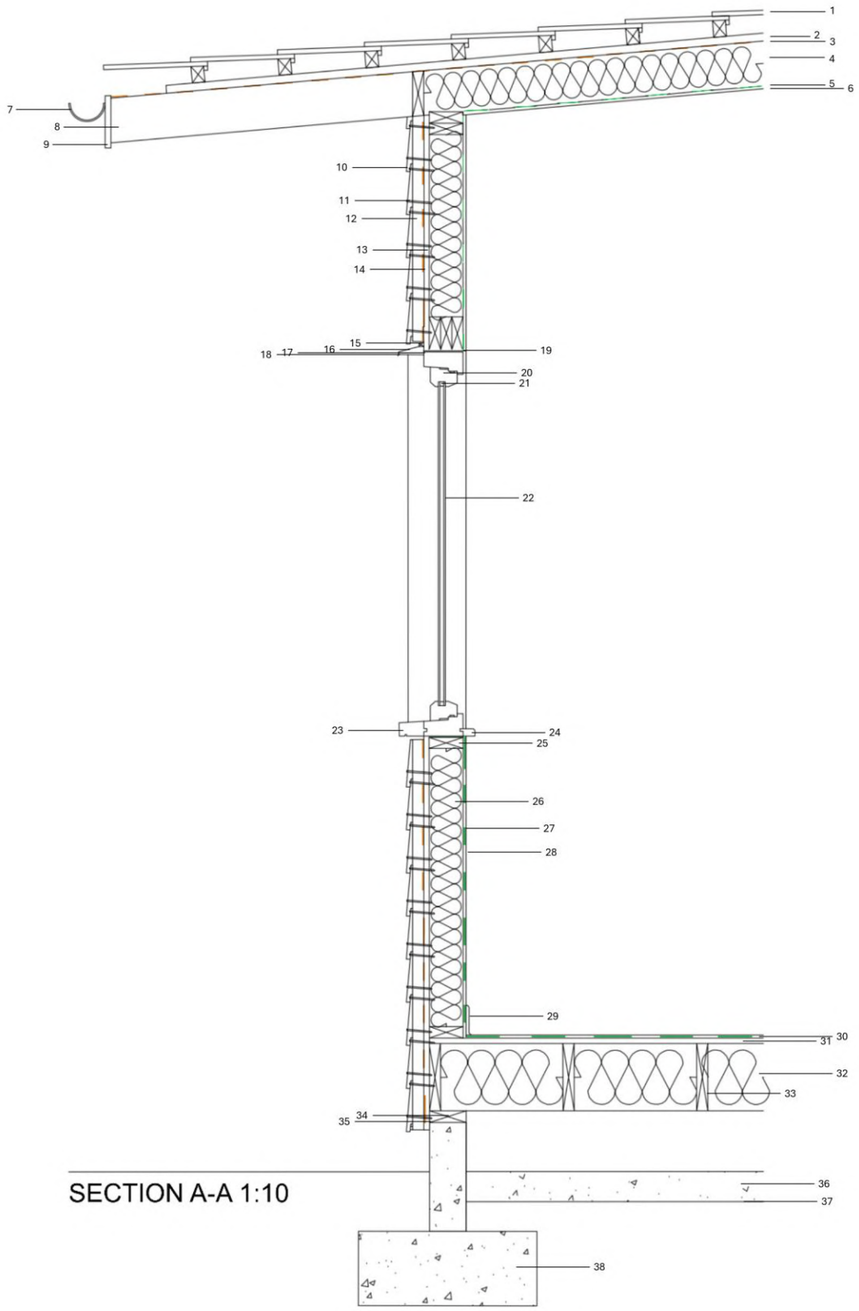
25mm x 50mm counter battens

Clay tiles fixed to counter battens

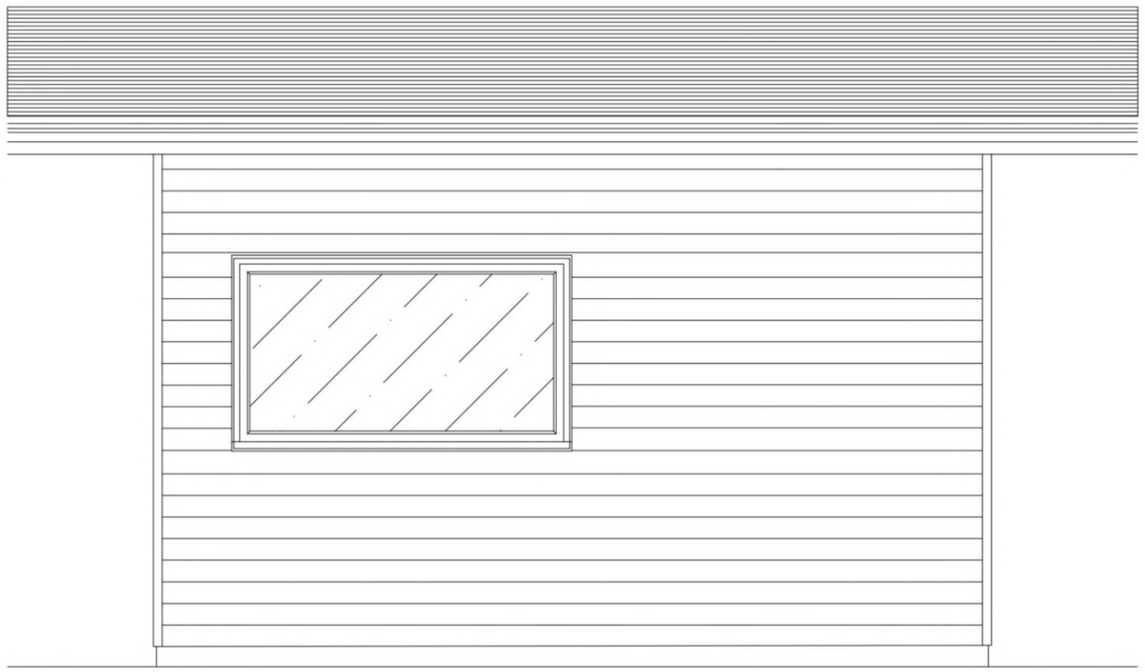


CH03 - DRAWN STUDY - PLAN, SECTION, ELEVATION





SECTION A-A 1:10



ELEVATION 1:20



PLAN 1:10

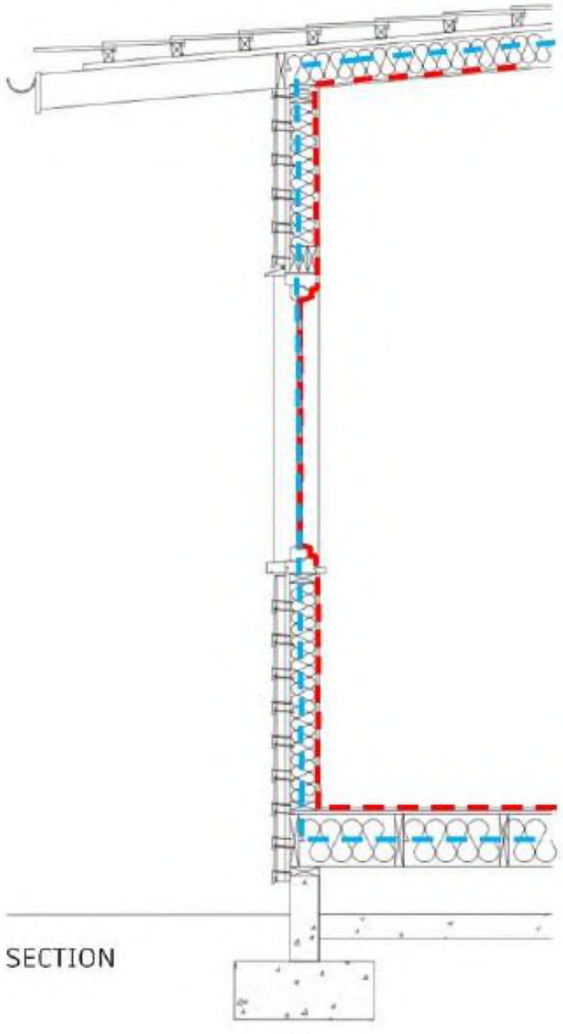
SECTION A-A

1 . Roof tiles on timber battens
2 . Counter timber batterns
3 . Breather membrane
4. Insulation
5 . Vapour control layer
6 . Plasterboard 10mm
7 . Gutter
8 . 38 x 152mm timber rafters
9 . Roof edge trim
1 0 . Fibre cement cladding
1 1 . Nails
1 2 . 25 x 38mm vertical timber battens
1 3 . Sheating 18mm
1 4 . Breather membrane
1 5 . Window Lintel
1 6 . Pre formed metal flashing
1 7 . Wood packing
1 8 . Timber casing with water drip
1 9 . Compressed foam sealing tape
2 0 . Timber window frame
2 1 . Spacer
2 2 . 24mm double glazed window
2 3 . Extended cill
2 4 . Internal cill
2 5 . 38 x 114mm timber studs
2 6 . Timber stud wall - insulation between studs
2 7 . Vapour control layer
2 8 . 10mm plasterboard
2 9 . Wall skirting
3 0 . 10mm timber flooring
3 1 . 18mm subfloor
3 2 . 228mm insulation between timber joist
3 3 . 38x228mm timber joists
3 4 . Sole plate
3 5 . Damp proof course
3 6 . Concrete slab
3 7 . Damp proof membrane
3 8 . Pad foundation

Vapour control layer

Breather membrane

CH04 - THERMAL AND FIRE PERFORMANCE AUDIT



SECTION



PLAN

Airtightness Line

Thermal Line

U-Values			
Pitched roof with insulation along rafters	Timber frame wall with timber cladding	Timber suspended floor	Passivhaus
			
0.18	0.3	0.17	0.10 - 0.15 W/m²K